AI / EMBODIED INTELLIGENCE / EMBEDDED SYSTEMS

Zhang Quanrui

BUPT / Telecommunications Engineering with Management / Class of 2029

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Study Room

Profile

Undergraduate at BUPT, Class of 2029, majoring in Telecommunications Engineering with Management. My first-semester average was 90.83/100, ranking in the top 6% of the major. I am building research and engineering depth across AI, embodied intelligence, and embedded systems through PRIS Lab research training, a sole-authored BiLSTM forecasting paper, and embedded software development for a blueberry-picking robot.

Education

BUPT / Telecommunications Engineering with Management

Class of 2029

B.Eng. Candidate / Sep 2025 - Jun 2029 (Expected)

Average 90.83/100 / Top 6% in major

- **Academics:** First-semester average of 90.83/100, ranking in the top 6% of the major.
- **Coursework:** Advanced Mathematics I (88), Linear Algebra (90), and Introduction to Computing and Programming (91).

Research Experience

Pattern Recognition and Intelligent Systems Lab (PRIS Lab)

Ongoing

Undergraduate Researcher / Advisor: Prof. Gao Shen

Embodied AI papers / Lab presentations / Robotics technical research

- **Focus:** Develop a connected understanding of agents and embodied intelligence, from model mechanisms to robotic system constraints.
- **Work:** Independently studied vision-language-action models including π0.5, presented findings in lab meetings, and evaluated robotic-arm, chassis, and embedded-control options for a blueberry-picking robot.
- **Outcome:** Built a repeatable workflow for paper decomposition, technical route comparison, and implementation-constraint analysis.

Short-term Bike-sharing Demand Forecasting with Bidirectional LSTM

Manuscript under review

Independent Researcher / Sole Author

Python / PyTorch / About 1M orders / MAE 55.50

- **Problem:** Built a two-layer bidirectional LSTM model for hourly urban bike-sharing demand forecasting.
- **Method:** Processed about one million Shenzhen open-data order records from Apr 2017 to Feb 2021, with hourly aggregation, cyclical time encoding, weekend indicators, and peak-hour features.
- **Result:** Multi-step outputs, Huber loss, and bidirectional modeling reduced MAE from 77.82 to 61.45; a random-forest fusion model further reduced it to 55.50, with RMSE 85.55.

BUPT-Tsinghua Embodied Intelligence Knowledge Base

Internal knowledge base

Core Contributor / University Embodied Intelligence Alliance

124 scholars / 122 labs / Structured academic resources

- **Role:** Owned information collection, manual verification, and structured entry for the Core Scholars and Core Labs modules.
- **Scale:** Completed structured records for 124 scholars and 122 laboratories, supporting a searchable and extensible academic resource base.
- **Boundary:** The knowledge base is internal; this public resume discloses only work scope and aggregate counts.

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Engineering Projects

Embedded Control System for a Blueberry-picking Robot Based on Renesas RA8P1

Project received a provincial second prize

Core Developer / Primary Embedded Software Contributor

C/C++ / RA8P1 / J-Link SWD / UART/CAN / FSM

- **Contribution:** In a three-person team, primarily owned embedded software architecture, core feature development, and hardware integration, while contributing to robotic-arm and chassis selection.
- **Implementation:** Implemented safety monitoring, robotic-arm UART communication, chassis motion control, gripper actuation, and an FSM-based picking workflow; debugged through J-Link/SWD, UART logs, and LED states.
- **Result boundary:** The project received a provincial second prize. The award belongs to the project outcome; personal contribution is stated separately because I was not a named certificate recipient.

NaviGoose Assistive Sensing System for the Visually Impaired

Approved under BUPT's 2025 Nestling Plan

AI Scene-recognition Design / Advisor: Assoc. Prof. Zhang Yang

LiDAR / 77GHz mmWave / Edge-cloud architecture

- **Contribution:** In a five-person team, owned AI scene-recognition planning and architecture analysis, and independently prepared the technical proposal, competitor research, and defense materials.
- **Approach:** Compared edge-only, cloud-only, and hybrid architectures, proposing edge-side real-time avoidance with cloud-side 3D reconstruction and path planning.

HarmonyOS Embedded App Development / Calculator

Completed

International School Starlight Project

HarmonyOS / Embedded application / Frontend interaction

- **Practice:** Designed and developed a basic calculator application, learning the HarmonyOS workflow, UI organization, and embedded-app debugging process.

Study Room / Immersive Web Study Space

Live

Independent Product and Frontend Development / GitHub Pages

React 19 / TypeScript / Vite / Zustand / PWA

- **Positioning:** Combines daily tasks, focus timers, ambient audio, study statistics, and completion feedback in one low-distraction interface.
- **Delivery:** Covers user scenarios, interaction states, responsive behavior, persistence, and GitHub Actions deployment.

Machine Learning Portfolio

Machine Learning and Deep Learning Notes: ML fundamentals, Neural networks and deep learning, Self-Attention, Transformer, BERT, Practice and review

Skills

Programming and Data	C/C++, Python; NumPy, Pandas, PyTorch, and Scikit-learn.
Models and Algorithms	LSTM, bidirectional RNNs, Transformers, and attention; continuing study of VLA models and embodied intelligence.
Embedded Systems	Renesas RA8P1, FSP, e2 studio, SWD/J-Link, UART, GPIO, FSMs, and hardware integration.
Research and Tooling	Academic writing, Zotero, literature search and review; Git, Linux, Anaconda, Markdown, and LaTeX.

Selected Awards and Activities

- CSP-J Olympiad in Informatics / Provincial Second Prize
- Lab4AI × BUPT AI-powered Academic Writing Workshop / Completion Certificate
- BUPT 2025 Nestling Plan / Approved Project Contributor
- Class Labor and Organization Committee Member
- International School Freshman Debate Tournament / Top 6